

COMBINED PROJECT: Advanced Web Architecture and Mobile Development

Document management and AI Chat based on RAG

Element	Value
Version	1.0 – functional and technical framing document
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Class	Engineer Cycle level 4, Computer Science and Information Systems track
Scope	iOS and Android mobile application, backend, PDF and image processing

Working document intended to define the design, development, testing and deployment of the product.

1. Executive Summary and Product Vision

The project involves designing a mobile conversation application with an AI Assistant capable of answering questions from a user-provided document corpus. The corpus mainly consists of PDF files and images; it may contain native text, scans, tables, diagrams, or photographs of documents.

The solution relies on a Retrieval-Augmented Generation (RAG) architecture. Documents are ingested, analyzed, segmented, and indexed so that, for a question, the most relevant passages are retrieved then transmitted to the language model (LLM). The answer must be explicitly linked to its evidence: each important statement is associated with one or more references opening the document, page, and, when possible, the excerpt or image area concerned. This approach revisits the principle of a verifiable answer highlighted by AI-assisted document search solutions [1] [2].

The product must also allow consulting previously generated conversations and their references without an Internet connection. The offline mode does not necessarily promise full generation of a new answer; it primarily guarantees access to the synchronized corpus, history, saved responses, and citations. A local generation extension could be considered as an option.

2. Context, Objectives, and Scope

2.1 Project Objectives

- Allow a user to import, organize and query a corpus consisting of PDFs and images.
- Provide concise, contextualized conversational responses with verifiable references.
- Reduce unfounded responses through a policy of replying only with elements found in the sources.
- Offer offline access to documents, conversations, responses and references previously synchronized.
- Prepare an augmented reality extension to display contextualized documentary instructions in the real environment.

2.2 Included Scope

Included in the MVP	Option
iOS/Android mobile application	Dedicated desktop version
Import PDFs and images from files, camera and gallery	Import of videos or web pages
Angular web portal for schools, users, roles, and corpus	Native desktop application
OCR of scanned documents and images	Advanced audio transcription
Hybrid text + vector indexing	Training or fine-tuning of a proprietary model
AI chat with citations and excerpts	Autonomous agents performing external actions
History and offline consultation	Full local generation on all devices
Multi-device synchronization	Advanced real-time collaboration

2.3 Constraints

The project assumes the existence of an API-accessible language model service and an embeddings service compatible with multilingual processing. Generation and initial indexing functions typically require connectivity, while offline viewing concerns content already downloaded. The rights to use imported documents remain the responsibility of the user or the organization deploying the solution.

3. Users and main journeys

Profile	Main needs	Value criterion
Student	Understand a corpus, ask questions, verify the cited passages	Time saving and traceability of the answers
Staff	Query procedures, standards or manuals internal	Operational and reliable answer
Field technician	Consult on-site instructions, sometimes without network	Fast access to the steps and warnings
Administrator / content owner	Manage corpora, rights, versions and metrics	Quality and safety mastery

3.1 Normal path: sourced question-answer

1. The user creates a document space and imports one or more PDFs or images.
2. The system displays the processing status: pending, OCR, segmentation, indexing, available or error.
3. The user opens a conversation and enters a question in natural language.
4. The engine retrieves relevant passages, generates an answer and associates citations with statements.
5. The user views the answer, opens a reference, views the source excerpt and can save or share the answer.

3.2 Offline path

When the device is offline, the user clearly sees the network state and the available perimeter. They can open documents marked as available offline, browse saved conversations, read previous answers, and access references. A new question can be saved in a local queue; if a local model is later selected, the system can offer a locally generated answer clearly marked as such.

4. Expected features

4.1 Management of spaces and sources

Feature	Description
Create a space	Name, description, main language, privacy settings.
Import PDFs	Selection from storage, system sharing or controlled download.
Import images	Gallery and camera; JPEG, PNG, HEIC formats to specify.
OCR	Extraction of text and preservation of coordinates of recognized blocks.
Metadata	Title, author, date, tags, version and processing status.
Delete / replace	Logical deletion

4.2 Conversation with the AI Assistant

Feature	Acceptance criteria
Ask a question	The user can send a question in text and check the status of generation.
Conversational context	Previous turns are taken into account within the configured limits. Of the more the responses concern only the schools to which the user is attached
Response with citations	Each response displays the references used as markers readable.
Source excerpt	A click opens the document to the relevant page or area, with highlighting if possible.
Response insufficiently supported by	The assistant indicates that it cannot find sufficient evidence at the place to invent.
Actions on response	Copy, save, share and report an incorrect response.

4.3 History and references

Each generated message must preserve the question, the answer, the date, the model used, the useful parameters, the retrieved passages and the displayed references. The reference sheet must present the document name, its version number, the page or image identifier, and the textual excerpt. References must remain accessible offline if the corresponding source has been synchronized.

4.4 Search in the corpus

A direct search, independent of the chat, should allow finding a document or passage by keywords, filters and semantic similarity. The engine will favor a hybrid search combining lexical and vector search, with reordering of results when it improves relevance [1].

4.5 Administrative Web Portal

A developed web portal complements the mobile application. It is intended for platform administrators, school managers and content managers. The portal centralizes school creation, account administration, role assignment, definition of documentary rights and monitoring of conversations or processes, without exposing administrative functions to mobile users.

Web Feature	Description
Dashboard	Overview of schools, users, corpus, processes and alerts.
School management	Create, modify, deactivate a school; manage its settings, managers and documentary spaces.
User management	Invite, search, modify, suspend and re-activate a user.
Role management	Create or modify roles and assign permissions and scopes to them documentary.
Assignments	Associate a user with a school, or a group and with one or more roles.
Document management	Associate sources with a school, a role, a class or a group; manage the versions.
Audit	Incorporate the principle of non-repudiation.

4.6 Authentication and account management

Primary authentication will be performed via the user's Google (Institutional) account, using an OAuth 2.0 / OpenID Connect flow. The portal and mobile app must use the same identity provider and the same account repository. After the first login, the backend verifies the token, links the Google identity to an internal account and applies the account, school, roles and permissions state. The term "Google Mail" is therefore treated as authentication with a Google account, with no password storage by the application.

Note: Creating Administrator accounts requires an email (gmail) and a password.

- Sign in with Google button on web and mobile.
- Server verification of the identity, issuer, audience and token expiration.
- Possibility to restrict access to Google Workspace domains authorized for a school.
- Creation of the internal profile: no document authorization should be granted solely because a Google account is valid.
- Logout, session revocation, management of suspended accounts and login tracing.

4.7 Document authorization by role

Mobile app users should receive only responses produced from documents within their authorization scope. Before any RAG retrieval, the backend computes access filters from the authenticated identity, the school, the groups and the active roles. These filters are applied at the search level and not only after generation, to prevent an unauthorized passage from entering the model context.

Role example	Possible documentary scope	Authorized response example
Administrator platform	All schools and all spaces according to delegation	Global reference framework or reports of administration
School administrator	A school and its documentary spaces	Internal regulations, procedures and school documents
Staff	Assigned school(s)	Courses, instructions and resources of its classes
Student / staff	Documents explicitly published for its school or its group	Response based solely on the accessible supports
Parent / guest	Explicitly shared documents	Public information or documents authorized

A role or association change must invalidate the affected caches and be reflected during the next mobile synchronization. The answers and references stored must preserve the authorization context and the document version at the time of their generation. If a document becomes inaccessible, the history may retain the answer according to the retention policy, but must indicate that the source is no longer available.

5. Functional and Technical Architecture

5.1 Logical view

Layer	Proposed components	Responsibilities
Web application	Angular 21 Framework	Administration of schools, users, roles, rights and corpus, Import
Mobile	iOS/Android, conversational UI, local storage (SQLite), synchronization. React Native Framework 0.87	Chat, visualization, cache, offline, synchronization
Backend	Node.js Framework (LTS version) or Nest.js (11.2.1)	Authentication, schools, users, roles, spaces, conversations, files, synchronization, API, control access
Pipeline documentary	Type detection, OCR, parsing, segmentation	Transform sources into indexable units
Search	Lexical index, vector base, reranker	Retrieve relevant passages
Generation	LLM, system prompt, guardrails	Produce a faithful and cited response
Observability	Logs, metrics, traces, feedback	Measure quality, costs, latency and errors

5.2 Target RAG Pipeline

6. Ingestion: file validation, antivirus, fingerprint, object storage and version creation.
7. Preprocessing: extraction of native text, OCR if needed, language detection and page/area identification.
8. Segmentation: cutting into passages with controlled overlap, preservation of provenance metadata.
9. Vectorization: creation of textual embeddings; possible extension to multimodal embeddings for images [1].
10. Retrieval: hybrid search, filtering by space and version, then reordering passages.
11. Generation: prompt requiring use of provided passages, citations and uncertainty wording.
12. Post-processing: citation validation, missing reference detection, response formatting and recording.

6. Non-functional requirements

Domain	Target requirement
Performance	Display the interface in under 2 seconds after opening; aim for a first response in under 8 seconds under nominal conditions.
Availability	The online service must be monitored and have a documented recovery strategy.
Scalability	Separate asynchronous ingestion/OCR processing from the conversational API.
Accessibility	Comply with WCAG principles applicable to mobile: contrast, dynamic sizes, screen reader and external keyboard navigation.
Deployment	Deploy the mobile application on the PlayStore.
Internationalization	Plan for French as the primary language and architecture for other languages.
Design	Component-oriented architecture, respected design pattern model

7. Offline mode and synchronization

7.1 Offline scope

Available offline	Internet-dependent in the MVP
Documents and images explicitly downloaded	Remote import and OCR server
Conversations and responses saved	Generation via remote LLM
References and associated extracts	Global reindexing of the corpus
Local search in synchronized content	Sharing to a remote space
Queue of questions to synchronize	Update of models and server parameters

7.2 Synchronization rules

- The user selects the spaces and sources available offline, with an estimate of the occupied size.
- Synchronization is differential and resumes after an interruption.
- The interface must never present as « offline » a response whose evidence is not available locally.

8. Augmented reality - feature (bonus)

The recommended use case is a ****contextualized document-based assistance on equipment or a location****. The user points the camera at a machine, a panel, an installation, a component, or an identified object. The application recognizes the context, queries the corresponding document corpus, and then overlays on the real view procedure steps, safety warnings, visual markers, and direct access to the source excerpts. This approach is consistent with the role of mobile AR: integrating digital information into the real environment using the camera and sensors [3]. It is also inspired by field support scenarios where guides, diagrams, and instructions are overlaid on the equipment [4].

8.1 Detailed scenario

13. The user selects a document space and enables « AR Assistance » mode.
14. The camera detects a QR/image marker or attempts markerless recognition, depending on the selected maturity level.
15. The system associates the object with a card or a set of versioned documents.
16. The RAG retrieves the passages corresponding to the request or the selected step.
17. The application displays 2D/3D markers, a short instruction, an optional warning, and a « View evidence » button opening the source page or area.
18. The user can confirm a step, take an annotated photo, and save the intervention in the history.

9. Functional Acceptance Criteria

- A native PDF, a scanned PDF, and an image containing text can be imported and identified.
- A question whose answer appears in the corpus produces a response accompanied by at least one correct reference.
- A question absent from the corpus triggers an expression of uncertainty and does not invent a reference.
- Opening a quote displays the correct document and the correct page or area, when the information is available.
- After synchronization, a user can view a response offline and all its downloaded references.
- Deleting a source prevents it from being retrieved in new responses.

10. Schedule and Deliverables

- Presentation of the projects according to the period from 04 to 08 January 2027. The presence of everyone is mandatory; the mobile application (Android) is deployed on PlayStore.
- Project management and source code are hosted on Github.
- **Strict application of design architecture models (MVC or HMVC, MVVM and Frontend Architecture).**
- **Strict application of component-oriented architecture**

11. References

[1] Google Cloud, « What is Retrieval-Augmented Generation (RAG)? », <https://cloud.google.com/use-cases/retrieval-augmented-generation>

[2] Google, « NotebookLM / Gemini Notebook – Understand Anything », <https://notebooklm.google/>

[3] IBM, « What is augmented reality? », <https://www.ibm.com/think/topics/augmented-reality>

[4] Salesforce, « What Is Augmented Reality in Field Service — and How it Can Transform Your Business », <https://www.salesforce.com/service/field-service-management/what-is-augmented-reality-in-field-service/>